PRODUCT USER MANUAL

For
OMI_CLIMATE_SI_BALTIC_extent

Issue: 4.0

Contributors : P. Lagemaa, U. Raudsepp, I. Maljutenko, L. Axell, V. Huess

Approval Date : November 2025

MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/09/06	all	Creation of the document	U. Raudsepp I. Maljutenko L. Axell V. Huess	P. Lagemaa
1.1	2019/01/21	all	Overall minor changes following OMI revision report	U. Raudsepp I. Maljutenko	P. Lagemaa
2.0	2019/10/10	all	Update to 2018 data	U. Raudsepp I. Maljutenko	P. Lagemaa
2.1	2021/10/01	all	Remove temporal extension specific details	P. Lagemaa	V. Huess
3.0	2023/09/01	all	Template change+ minor changes	P. Lagemaa	V. Huess
4.0	2025/05/22	all	Template change+ minor changes	P. Lagemaa	V. Huess

TABLE OF CONTENTS

RECORD TABLE **2**

GLOSSARY AND ABBREVIATIONS **4**

DATA ACCESS **5**

1) INTRODUCTION **6**

a) Summary 6

2) DESCRIPTION OF THE PRODUCT SPECIFICATION **7**

a) General Information 7

b) Production System Description 8

c) Dataset details 8

3) FILE FORMAT **9**

4) FILES NOMENCLATURE **9**

5) REFERENCES **9**

GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
SIE	Sea Ice Extent

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the Copernicus Marine Service Ocean Monitoring Indicator (OMI) OMI_CLIMATE_SI_BALTIC_extent. It contains two datasets:

- omi_climate_si_baltic_extent
- omi_climate_si_baltic_extent_yday

The OMI is derived from daily sea ice concentration products SEAICE_BAL_PHY_L4_MY_011_019 and SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004. The data is daily time series of the ice extent in the Baltic Sea derived from the operational ice charts.

Ice charts are produced by manually drawing polygons of ice, where each polygon has a set of attributes, including sea ice concentration and level ice thickness. The data used to produce the charts are satellite images (SAR) as well as in situ observations (reports) from ice breakers in the Baltic Sea. The ice charts are produced almost daily during the ice season and cover the period November 1979 to present.

Daily SIE values are computed from the daily sea ice concentration (SIC) maps. All sea ice covered ocean is included, lake ice is not. SIE is computed as the sum of the area of the grid cells that have a SIC value larger than 15%.

The product is published on the Copernicus Marine Service dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the Copernicus Marine Service Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The OMI is further described in the corresponding Quality Information Document (QUID).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_CLIMATE_SI_BALTIC_extent
Geographical coverage	Baltic Sea
Variables	Sea ice extent
Available time series	From 1993 to Month-4
Temporal resolution	Daily
Horizontal resolution	Integrated
Integration depth (for example)	surface
Format	NetCDF4

b) Production System Description

The sea ice extent (SIE in km2) time series have been derived from daily sea ice charts from SEAICE_BAL_PHY_L4_MY_011_019 and SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004. The product provides SIE for the Baltic Sea from January 1980 until present. Weekly smoothed time series of the SIE have been calculated from daily values using moving average filter with 7-day window.

The annual course of the SIE has been calculated as daily mean ice extent for each day-of-year in the period of 1993-2022. Same way the standard deviation is calculated from ice areas occurring during each of the day-of-year.

The SIE from the time period of October to September is used to show ice extent during the last winter.

c) Dataset details

omi_climate_si_baltic_extent (daily sea ice extent time series smoothed over 7-day time period) contains the following variables:
siextentn [10e3 km2] long_name: Sea Ice Extent standard_name: sea_ice_extent **time** [days since 1950-01-01 00:00:00] long_name: time standard_name: time
omi_climate_si_baltic_extent_yday (time series of mean year day sea ice extent) contains the following variables:
siextentn_yday_mean [10e3 km2] long_name: Year Day Average Sea Ice Extent standard_name: sea_ice_extent **siextentn_yday_std** [10e3 km2] long_name: Year Day Sea Ice Extent Standard Deviation standard_name*: sea_ice_extent **siextentn_yday_yyyy** [10e3 km2] long_name: Sea Ice Extent of yyyy-1/yyyy standard_name*: sea_ice_extent **time** [days since 1950-01-01 00:00:00] long_name: time standard_name: time

* The standard name is defined closest available in CF conventions yet it does not reflect the content of the variable. Please use variable name as a reference to the content.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

Files nomenclature when using the get command in the Copernicus Marine Toolbox or using the Browse File is as follows:

<datasetname>_P<productiondate>.nc

Where :
<datasetname> is the dataset reference
<productiondate> is the production date when the file was generated
.nc is the standard NetCDF filename extension.

5) REFERENCES

Quality Information Document (CMEMS-BAL-OMI-QUID-CLIMATE-SI-extent)